

Economic Prosperity & Football Success:

Turkey & England (1993–2024)

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1. Overview

This project investigates whether a nation's macroeconomic conditions — exchange rates, GDP growth, and CPI inflation — meaningfully predict the international football performance of its clubs and national team. The central question is motivated by a striking historical observation: Turkey's greatest football achievements — Galatasaray's 2000 UEFA Cup and Super Cup, Beşiktaş's unbeaten 2017–18 Champions League group stage, and the national team's 3rd-place finish at the 2002 World Cup — all coincided with periods of relative economic stability. The study covers 31 years (1993–2024), encompassing two countries (Turkey and England), five Turkish clubs, six English Premier League clubs, and both national teams.

The analysis was structured across four phases. First, exploratory data analysis (EDA) to understand the individual and joint behaviour of economic and football time series. Second, formal Pearson and Spearman hypothesis testing to assess whether economic health is statistically associated with UEFA coefficient performance and FIFA rankings. Third, OLS regression to quantify the joint explanatory power of economic variables. Fourth, machine learning experiments using K-Means clustering, PCA, and three supervised classifiers to test whether economic features can reliably classify high- versus low-football-performance periods.

The central finding is that Turkey's club-level UEFA performance is significantly and positively correlated with economic health ($r = +0.225$, $p = 0.037$), and that macroeconomic variables jointly explain 60.9% of the variance in Turkish club UEFA points (OLS $R^2 = 0.609$, $F = 31.13$, $p < 0.001$). The strongest single finding is a large negative correlation between inflation and club success ($r = -0.590$): Turkey's hyperinflation decades (1994–2004) coincided almost exactly with its worst European football results. England's clubs, decoupled from domestic economics by global broadcast revenues and foreign ownership, show no such relationship.

2. Motivation

Turkish football has oscillated between historic highs and prolonged mediocrity in European competition, often in ways that correlate visually with the country's economic trajectory. Three landmark moments define Turkish football's European legacy and frame the central hypothesis of this project:

- Galatasaray (2000): Won the UEFA Cup (defeating Arsenal on penalties) and the UEFA Super Cup (defeating Real Madrid 2–1). This remains the only major European trophy won by a Turkish club. Turkey's inflation rate that year was 55% — elevated but sharply declining from the 80–100% levels of the early 1990s.
- Beşiktaş (2017–18 Champions League): Became the first and only Turkish club to top their Champions League group stage, finishing above Porto, Monaco, and RB Leipzig without a single defeat. This occurred during a period of moderate (by Turkish standards) economic conditions before the 2018 currency crisis.
- Turkey National Team (2002 World Cup): Finished 3rd in the FIFA World Cup, defeating co-host South Korea and Brazil's qualification hopes — the highest achievement in Turkish national team history. The early 2000s represented a brief window of relative macroeconomic stabilisation before the 2001 banking crisis shock.

England provides a deliberate structural counterpoint. The Premier League's global broadcast deals (£5.1 billion domestic rights for 2022–25), Champions League prize money (€100M+ per deep run), and foreign ownership (Abu Dhabi's acquisition of Manchester City in 2008, Qatar's acquisition of PSG influencing the broader ecosystem) mean that English club success is largely insulated from domestic GDP or inflation fluctuations. If the Turkish correlation is real and not an artefact, we would expect English clubs to show no such relationship — and indeed they do not.

Research Hypotheses

H₁ (Alternative Hypothesis): Stronger economic conditions are positively associated with better international football performance ($\rho > 0$, one-tailed Pearson/Spearman test, $\alpha = 0.05$).

H₀ (Null Hypothesis): There is no statistically significant linear correlation between economic indicators and football success ($\rho = 0$).

The one-tailed formulation is justified by the directional a priori hypothesis: we expect better economics → better football, not the reverse. A two-tailed test would be more conservative; we report both one- and two-tailed p-values where relevant for transparency.

3. Data Sources & Collection

Each country contributes 64 half-year observations (H1: January–June; H2: July–December), yielding a combined dataset of 128 rows used for machine learning. The half-year granularity was chosen to capture within-year variation in UEFA competitions, which run September–May (H2 of year t and H1 of year $t+1$), while remaining compatible with the annual GDP and CPI data available from primary sources.

Turkey — 64 Observations (1993–2024)

- Exchange Rate: USD/TRY monthly average, sourced from the Central Bank of Turkey (TCMB Electronic Data Delivery System, evds2.tcmb.gov.tr). Half-year averages computed from monthly observations.
- GDP Growth Rate: Annual real GDP growth (% year-on-year), sourced from World Bank Open Data (indicator NY.GDP.MKTP.KD.ZG). Assigned to both H1 and H2 of the same year.
- CPI Inflation: Annual CPI inflation rate (%), sourced from World Bank / TurkStat (indicator FP.CPI.TOTL.ZG).
- Club UEFA Coefficients: Half-year UEFA coefficient points for Galatasaray, Fenerbahçe, Beşiktaş, and Trabzonspor, reconstructed from UEFA.com and kassiesa.net/uefa/coeff. Points reflect European match results (wins, draws, group stage advancement, knockout rounds).
- National Team FIFA Ranking: Bi-annual FIFA World Ranking snapshots (January and July), sourced from FIFA Historical Archive (fifa.com/fifa-world-ranking). Converted to a FIFA Index where lower rank = higher score.

England — 64 Observations (1993–2024)

- Exchange Rate: GBP/USD monthly average, sourced from the Bank of England Statistical Database (bankofengland.co.uk). For England, a stronger pound = higher index score (no inversion needed).
- GDP Growth & CPI Inflation: World Bank Open Data (indicators for GB).
- Club UEFA Coefficients: Manchester City, Manchester United, Arsenal, Liverpool, Tottenham Hotspur, Chelsea — the Premier League 'Big 6' throughout most of the study period.
- National Team FIFA Ranking: Same methodology as Turkey.

Composite Economic Health Index

A composite Economic Health Index (EHI, range 0–1, higher = better economic conditions) was constructed to serve as the primary economic predictor. The formula combines three normalised and directionally consistent indicators:

- Step 1 — Min-max normalise each variable across the full 64-period series: $x_{\text{norm}} = (x - \min) / (\max - \min)$
- Step 2 — Invert exchange rate for Turkey ($1 - \text{norm_rate}$), since weaker lira = worse economic conditions
- Step 3 — Invert inflation ($1 - \text{norm_cpi}$), since higher inflation = worse economic conditions
- Step 4 — Average the three normalised components: $\text{EHI} = (\text{norm_rate_adj} + \text{norm_gdp} + \text{norm_cpi_adj}) / 3$

For England, the GBP/USD rate is used directly without inversion (stronger pound = higher EHI). GDP growth and inflation are treated identically to Turkey.

Data Quality & Limitations

- GDP and CPI are annual figures assigned to both H1 and H2 of each year, reducing temporal granularity. Monthly versions would improve the analysis but were not available in the same format.
- Club UEFA coefficient points for individual half-years were reconstructed from archival data. While cross-validated against multiple sources, exact values should be treated as estimates.
- The USD/TRY series spans three distinct monetary eras: pre-2001 (old lira, 10^6 scale), 2001–2004 (transition), and post-2005 (new lira). A log transformation was applied before normalisation to handle the extreme scale differences.

4. Visualizations & Exploratory Data Analysis

The EDA phase generated six categories of visualisation for each country. Figures are referenced below as they would appear in the analysis notebook (analysis_v3.py) and project webpage.

Economic Time Series — Turkey

The first set of figures shows the three raw economic indicators for Turkey over 1993–2024, with the composite EHI overlaid. Several periods stand out immediately. The USD/TRY series is the most visually dramatic: it ranges from approximately 0.15 in 1993 to over 27 in 2024, with particularly sharp depreciations in 1994 (banking crisis), 2001 (banking system collapse), 2018 (currency crisis, –28% in one year), and 2021–2022 (monetary policy controversy, –44% in 2021 alone). The CPI inflation series mirrors this: rates exceeding 100% in 1994 and 1997, declining to single digits in 2004–2010, then rising sharply again from 2017 onward, reaching 85% in 2022.

The composite EHI therefore has two clear peaks: the early 2000s stabilisation period (2002–2007, corresponding to EU accession negotiations and IMF stand-by agreements) and a brief 2009–2013 window of moderate conditions. These are precisely the eras when Turkish football was performing best in Europe — a visual correlation that the statistical tests subsequently formalise.

Figure 1. USD/TRY exchange rate (top), CPI inflation (middle), and composite Economic Health Index (bottom) for Turkey, 1993–2024, with half-year granularity.

Football Performance Time Series — Turkey

The Turkish club UEFA coefficient series shows a pronounced peak around 2000–2003, driven by Galatasaray's dominant runs (5–7 pts/half) alongside consistent Fenerbahçe and Beşiktaş participation. A secondary peak appears in 2016–2018, anchored by Beşiktaş's 2017–18 UCL group stage. The 1993–1997 period shows very low UEFA points despite some club participation, consistent with Turkey's hyperinflation era reducing clubs' ability to retain and recruit European-quality talent.

The national team FIFA index shows Turkey's all-time high rank around 2002–2004 (reaching approximately rank 5 globally following the World Cup 3rd place), then a prolonged decline to the 20–40 range as the generation of Hakan Şükür, Rüştü Reçber, and Emre Belözoğlu aged out without adequate replacements.

Figure 2. Total Turkish club UEFA coefficient points per half-year (top) and national team FIFA ranking index (bottom), with key historical milestones annotated: GS 2000 UEFA Cup, Turkey 2002 WC, BJK 2017–18 UCL.

Economic Time Series — England

England's economic indicators tell a structurally different story. The GBP/USD rate oscillates in a relatively narrow band (1.35–2.10) compared to Turkey's multi-order-of-magnitude changes. GDP growth is consistently positive except for the 2008–09 Global Financial Crisis (–4.2%) and 2020 COVID shock (–9.8%). CPI inflation is low and stable throughout: 1–4% in most years, with only the 2022–23 post-COVID supply shock pushing it above 10%.

The English EHI therefore lacks the dramatic volatility of the Turkish series — it is a relatively flat, high-value series with two dips (2008–09 GFC, 2020). This structural difference is itself an important finding, as it explains why the ML models can perfectly classify country of origin from economic features alone.

Figure 3. GBP/USD exchange rate, CPI inflation, and composite Economic Health Index for England, 1993–2024.

Football Performance Time Series — England

English club UEFA points follow a strong upward trend across the full period, reflecting Premier League dominance in European competition from the late 1990s onward. Manchester United's 1999 treble, Arsenal's 2005–06 CL final, Chelsea's 2012 CL title, and Manchester City's 2019–2023 dominance all contribute to this trajectory. Crucially, this upward trend is structurally similar to what would be expected from increasing broadcast revenue — not from macroeconomic conditions. The negative (non-significant) correlation between England's EHI and club UEFA points ($r = -0.398$) is consistent with an omitted variable: the real driver is private investment and broadcast revenue, not the domestic economy.

Figure 4. Total English club (Big 6) UEFA coefficient points per half-year, 1993–2024. Annotated: Man Utd treble 1999, Arsenal CL final 2006, Man City Abu Dhabi acquisition 2008.

Dual-Axis Overlay: Economic Health Index vs. Club UEFA Points

The centrepiece visualisation of the EDA phase overlays the composite EHI against total club UEFA points for Turkey on a single timeline. The visual co-movement is clear in the 1999–2007 window — EHI rises as inflation falls, and Turkish clubs simultaneously achieve their best European results. The 2016–2018 secondary EHI mini-peak also aligns with Beşiktaş's UCL group stage success. The 2022–2024 period shows both series declining together: Turkey's renewed inflation crisis and the near-absence of Turkish clubs in UEFA group stages.

For England, the overlay shows divergence: EHI is relatively flat while UEFA points trend strongly upward. This visual divergence is the qualitative signal that formal hypothesis testing subsequently confirms.

Figure 5. Dual-axis overlay of composite EHI and total club UEFA points for Turkey (left panel) and England (right panel), 1993–2024.

Distribution Analysis

Distribution plots reveal two important properties relevant to test selection. The total club UEFA points variable is strongly right-skewed for Turkey (skewness ≈ 1.78): the vast majority of half-years have low UEFA participation (0–5 points), with rare peaks of 15–20+ points when Galatasaray or Beşiktaş made deep runs. The D'Agostino K^2 normality test rejected normality for both the economic and football variables (all $p < 0.01$), which motivated the inclusion of non-parametric Spearman ρ alongside Pearson r as a robustness check. Both tests produced directionally consistent results, and Spearman coefficients were within 0.03–0.05 of Pearson values, confirming the correlations are not artefacts of distributional assumptions.

Figure 6. Empirical distributions of composite EHI (left), total club UEFA points (centre), and CPI inflation (right) for Turkey, with KDE overlays. All three variables failed the D'Agostino K^2 normality test.

Per-Club EHI Scatter Plots

Individual scatter plots of EHI versus per-club UEFA points reveal heterogeneity across clubs. Beşiktaş shows the strongest correlation ($r = +0.312$, $p = 0.006$), driven by their strong results in the 2016–2018 window coinciding with moderate economic conditions. Fenerbahçe also shows a significant positive relationship ($r = +0.254$, $p = 0.021$). Galatasaray is close to the boundary of significance ($r = +0.198$, $p = 0.058$) — their dominant 2000 run occurred during a particular economic transition point that the EHI index captures only partially. Trabzonspor shows near-zero correlation, consistent with their limited European participation across the study period.

Figure 7. EHI vs. per-club UEFA points scatter plots for Galatasaray, Fenerbahçe, Beşiktaş, and Trabzonspor with linear trend lines and 95% confidence bands.

5. Hypothesis Testing

Nine Pearson correlation tests were conducted per country, using one-tailed tests ($H_1: \rho > 0$) at $\alpha = 0.05$ with 95% Fisher z -transformed confidence intervals. Spearman ρ was computed as a non-parametric backup for all primary tests; results were consistent throughout. Normality was pre-checked using D'Agostino K^2 (all variables rejected normality, n.s. Spearman confirms robustness).

Full Results Table

Country	Variable Pair	r	p (1-tail)	R ²	Effect	H ₀
🇹🇷 TR	Economic Index × Total Club UEFA	+0.225	0.037	5.1%	Small (+)	REJECTE D ✓
🇹🇷 TR	Economic Index × National Team FIFA	+0.166	0.096	2.7%	Small (+)	KEPT
🇹🇷 TR	USD/TRY × Total Club UEFA	+0.428	0.0002	18.3 %	Medium (+)	REJECTE D ✓
🇹🇷 TR	CPI Inflation × Total Club UEFA	-0.590	≈1.000	34.8 %	Large (-)	KEPT (neg. dir.)
🇹🇷 TR	Economic Index × Galatasaray UEFA	+0.198	0.058	3.9%	Small (+)	KEPT (border)
🇹🇷 TR	Economic Index × Beşiktaş UEFA	+0.312	0.006	9.7%	Medium (+)	REJECTE D ✓
🇹🇷 TR	Economic Index × Fenerbahçe UEFA	+0.254	0.021	6.4%	Small (+)	REJECTE D ✓
🇬🇧 ENG	Economic Index × Total Club UEFA	-0.398	0.999	15.8 %	Medium (-)	KEPT (neg. dir.)
🇬🇧 ENG	Economic Index × National Team FIFA	+0.240	0.028	5.8%	Small (+)	REJECTE D ✓

Note on $p \approx 1.000$ entries: For one-tailed tests with $H_1: \rho > 0$, a negative correlation produces $p = P(T > t_{\text{obs}})$ where t_{obs} is strongly negative. For $r = -0.590$ ($n = 64$), $t = -5.75$, and $p = 1 - \Phi(-5.75) \approx 1.000$. This does not mean the correlation is non-significant in the two-tailed sense (two-tailed $p < 0.0001$) — it means the effect is significant in the opposite direction from H_1 . The KEPT label reflects that H_1 (positive direction) is not supported.

Key Findings — Turkey

H₁ Supported at Club Level (Economic Index × Total Club UEFA)

The primary hypothesis is supported for Turkish clubs: $r = +0.225$ (95% CI: +0.03 to +0.40), $p = 0.037$, $R^2 = 5.1\%$. While the bivariate effect size is small (Cohen's r threshold for 'small' is 0.10, 'medium' is 0.30), it is statistically significant and in the predicted direction. The effect becomes substantially stronger when economic components are disaggregated (see inflation results below).

Inflation as the Dominant Single Predictor

The most striking result is the large negative correlation between CPI inflation and club UEFA points: $r = -0.590$, $R^2 = 34.8\%$. This is a large effect by any standard. The one-tailed p is ≈ 1.000 because H_1 specified a positive direction — but the two-tailed p is < 0.0001 . This is not contradictory: it means inflation is a highly significant predictor, but in the negative direction. High inflation destroys club purchasing power for foreign transfer fees and salaries, denominated primarily in euros or dollars, while club revenues remain in lira. The 1990s hyper-inflation decade (1994–2001, average CPI = 78%) corresponds almost exactly to Turkey's worst decade of European football results.

USD/TRY — Spurious Positive Direction

The positive $r = +0.428$ for USD/TRY $\times$ Club UEFA ($p = 0.0002$) is counterintuitive: a weaker lira correlates with better club results? This is a spurious correlation driven by a shared long-run time trend. Both the USD/TRY rate and cumulative UEFA coefficients have trended upward over the 1993–2024 period — the exchange rate due to persistent inflation differentials, and UEFA points due to expanded European competition formats and more Turkish club participation. The composite EHI corrects for this by inverting the exchange rate and normalising, which is why EHI $\times$ Club UEFA gives $r = +0.225$ (correct direction) while raw USD/TRY $\times$ Club UEFA gives $r = +0.428$ (spurious). This is analogous to Ali Çaylak's raw $r = +0.49$ (spurious) vs. stationary $r = -0.02$ finding in the arXiv/BOTZ project.

Key Findings — England

Clubs Decoupled from Domestic Economy

The negative correlation between England's EHI and club UEFA points ($r = -0.398$) is the mirror image of the Turkish result — and equally informative. English clubs perform better in European competition during periods of domestic economic weakness (2008–09 GFC, 2020 COVID). This reflects the 'arms race' dynamics of the Premier League: foreign capital inflows (Abu Dhabi's Man City purchase in August 2008 — three weeks after the GFC began — being the canonical example) are largely uncorrelated with UK GDP or inflation. The Global Financial Crisis reduced real estate and commodity prices but simultaneously opened England to a wave of foreign club acquisitions.

National Team Shows Weak Economic Link

England's national team does show a small but significant positive correlation with EHI ($r = +0.240$, $p = 0.028$). This is consistent with the idea that domestic economic conditions influence grassroots football development and youth academies even when top club performance is insulated. However, the R^2 is only 5.8%, and confounders (tournament format changes, qualification paths) make causal inference difficult.

6. OLS Regression Analysis

Ordinary Least Squares regression was applied to the Turkish dataset ($n = 64$), regressing total club UEFA points per half-year on three economic predictors simultaneously: Economic Health Index, GDP growth rate, and CPI inflation. This addresses the limitation of pairwise correlations by quantifying the joint explanatory power of economic variables while controlling for their intercorrelations.

Model Performance

- $R^2 = 0.609$ — economic variables jointly explain 60.9% of the variance in Turkish club UEFA performance
- Adjusted $R^2 = 0.589$ — adjusted for the three predictors, confirming the model is not over-parameterised
- F-statistic = 31.13, $p < 0.001$ — the model as a whole is highly statistically significant

- Residual standard error = 3.84 UEFA points — typical prediction error within ± 4 points per half-year

Coefficient Table

Predictor	Coefficient (β)	Std. Error	t-stat	p-value
Economic Health Index	+0.813	0.147	+5.53	< 0.001
GDP Growth Rate	+0.291	0.112	+2.60	0.012
CPI Inflation	-0.447	0.098	-4.56	< 0.001
Intercept	+12.34	3.21	+3.84	< 0.001

Interpretation

All three predictors are significant at $p < 0.05$. The Economic Health Index has the largest standardised coefficient, confirming its role as the primary composite predictor. CPI Inflation has a large negative coefficient ($\beta = -0.447$): holding other variables constant, each one-unit increase in normalised inflation is associated with a 0.45-point decrease in half-year UEFA points. GDP Growth has a positive effect ($\beta = +0.291$): periods of economic expansion, controlling for inflation and the exchange rate, are associated with better European results.

The high R^2 (60.9%) is a strong result for macroeconomic predictors in a sports context. For comparison, a naive model predicting the mean UAE performance for every period would have $R^2 = 0$. The OLS model substantially outperforms this baseline. The result implies that approximately 39% of variance in Turkish club UEFA performance is attributable to non-economic factors: managerial quality, transfer luck, club governance, and player development.

Residual Diagnostics

Residual plots show approximate homoscedasticity and no clear autocorrelation pattern. The Breusch-Pagan test for heteroscedasticity did not reject the null ($p = 0.18$). A Durbin-Watson statistic of 1.94 indicates no significant first-order autocorrelation in the residuals. Leverage analysis identified two periods (H2 1994 and H1 2001) as moderate outliers corresponding to the 1994 banking crisis and 2001 IMF intervention — both extreme economic events that produced atypical football outcomes.

Figure 8. OLS residual diagnostics: residuals vs. fitted (top left), Q-Q plot (top right), scale-location (bottom left), leverage plot (bottom right).

7. Machine Learning Analysis

The machine learning phase asks whether economic features can reliably classify half-year periods as 'high' or 'low' football performance using supervised classifiers, and whether unsupervised methods identify economically meaningful eras in the data. The target variable for

supervised tasks was a binary label: periods where total club UEFA points exceeded the median were labelled 'High Performance' (1); below-median periods were labelled 'Low Performance' (0). Five-fold stratified cross-validation was used throughout to ensure robust accuracy estimates.

Feature Set

- Economic Health Index (composite, 0–1 scale)
- USD/TRY or GBP/USD exchange rate (half-year average)
- GDP growth rate (annual, assigned to both halves)
- CPI inflation rate (annual, assigned to both halves)
- Country indicator (0 = Turkey, 1 = England) — for combined dataset only

Four models were trained per dataset: Logistic Regression (L2 regularisation, C optimised via CV), Random Forest (100 trees, max_depth via CV), and Gradient Boosting (learning rate and n_estimators via CV). A majority-class baseline was also computed for each dataset.

Cross-Validation Accuracy Results

Dataset	Log. Reg.	Rand. Forest	Grad. Boost	Baseline
🇹🇷 Turkey — 64 rows	0.674	0.688	0.688	0.609
🇬🇧 England — 64 rows	0.806	0.860	0.845	0.594
⚽ Combined — 128 rows	0.851	0.907*	0.891	0.516
🌐 Country ID (TR vs ENG)	1.000	1.000	1.000	0.500

* Best result: Random Forest on combined 128-row dataset, CV accuracy = 90.7%

Turkey — 64 Rows

With 64 observations and a majority-class baseline of 60.9% (slightly imbalanced: more low-performance periods), all three models outperform the baseline. The best accuracy is 68.9% (Random Forest and Gradient Boosting tied). While modest, this is meaningful: economic features alone correctly classify 69% of half-years, compared to a 61% baseline for always predicting the majority class. The improvement is modest because $N = 64$ is small for reliable CV, and because several non-economic factors (Hakan Şükür's generation, the 2008 Fenerbahçe match-fixing scandal, club boardroom instability) add noise.

Figure 9. Turkey-only classifier results: confusion matrix for Random Forest (left) and feature importances (right). Inflation and EHI are the top two predictors.

England — 64 Rows

The 86.0% accuracy for England's Random Forest classifier initially appears surprising given the negative (non-significant) correlation between England's EHI and club performance. The explanation is that England's model is primarily learning the time trend in both series: economic

conditions (especially the 2008 GFC dip and recovery) and club UEFA performance both have temporal structure that a classifier can exploit. The country indicator in the combined model further amplifies this: the structural differences between Turkish and English economic regimes are so pronounced that classifying country of origin achieves 100% accuracy.

Combined Dataset — 128 Rows

Combining Turkey and England to 128 rows, as suggested by the course instructor (Neşe Hoca) to address the small sample concern, raises the best accuracy to 90.7%. The combined model has twice the training data and benefits from the structural contrast between the two countries. The country indicator variable alone carries substantial discriminative power, which is why the combined accuracy substantially exceeds the individual-country models.

Country Identification — 100% Accuracy

A binary classifier trained to predict country of origin (Turkey vs. England) from economic and football features alone achieves 100% cross-validated accuracy with all three classifiers. This confirms what the EDA already suggested: Turkey and England occupy completely non-overlapping regions of the economic-football feature space. A Turkish half-year with $EH I > 0.7$ and Club UEFA points > 12 does not exist in the data; an English half-year with $EH I < 0.3$ and Club UEFA points > 20 does not exist either. The two countries are structurally orthogonal in this feature space.

Figure 10. PCA projection of the combined 128-row dataset. Turkey (red) and England (blue) occupy completely separate regions of the 2D PCA space, confirming 100% classification accuracy.

K-Means Clustering (k = 3, Elbow Method)

K-Means was applied to the Turkish 64-row dataset with features: EHI, CPI inflation, USD/TRY, and total club UEFA points. The elbow method (within-cluster sum of squares vs. k) identified $k = 3$ as the optimal number of clusters. PCA explained 57% of total variance in the first two components, enabling a 2D cluster visualisation.

Cluster / Era	N (periods)	UEFA Performance	Economic Profile
Cluster 0 — 1990s Hyper-Inflation Era	~18 periods	Low (avg 3.2 pts/half)	Inflation $> 40\%$, USD/TRY < 0.5
Cluster 1 — 2000s Stability Era	~22 periods	Peak (avg 8.7 pts/half)	Lower inflation, stable currency
Cluster 2 — 2010s–2020s Mixed Era	~24 periods	Moderate (avg 5.4 pts/half)	Currency depreciation, moderate inflation

Figure 11. K-Means clustering (k=3) of Turkish half-year periods: PCA 2D projection coloured by cluster (top), and economic/football profiles per cluster (bottom). The three clusters map naturally onto Turkey's economic-footballing eras.

The three clusters map almost perfectly onto historically recognised economic eras in Turkey. Cluster 0 (1990s hyperinflation era) captures the devastating combination of 60–100% inflation, a rapidly depreciating lira, and weak European club results. Cluster 1 (2000s stability era) corresponds to the IMF stand-by agreement period, EU accession optimism, and Turkey's peak

football achievements. Cluster 2 captures the mixed 2010s–2020s: currency depreciation accelerating, inflation rising again, but club UEFA performance more variable due to individual club management quality.

Feature Importances

Across all Random Forest models, the most important features for predicting high/low club performance are consistently CPI inflation (importance ≈ 0.38) and the composite EHI (importance ≈ 0.31), followed by GDP growth (≈ 0.19) and exchange rate (≈ 0.12). This mirrors the OLS coefficient magnitudes and confirms that inflation is the primary macroeconomic driver of Turkish football performance — a finding that holds across both parametric (OLS) and non-parametric (Random Forest) modelling frameworks.

Figure 12. Random Forest feature importances for Turkey-only classifier. CPI inflation and composite EHI are the dominant predictors.

8. Conclusions

This project set out to answer one question: does a nation's economic health predict its football clubs' and national team's international performance? The answer is a qualified yes for Turkey and a no for England — and this divergence is itself the most informative finding.

Turkey — Economy Matters

- H_1 is supported at the club level: $r = +0.225$, $p = 0.037$. A stronger Turkish economy is statistically associated with better club UEFA performance.
- Macroeconomic variables jointly explain 60.9% of the variance in Turkish club UEFA points (OLS $R^2 = 0.609$, $F = 31.13$, $p < 0.001$). This is a strong result for a purely macroeconomic model of sports performance.
- Inflation is the single strongest driver: $r = -0.590$. The 1990s hyperinflation decade (average CPI = 78%) corresponds almost exactly to Turkey's worst European club results. When inflation fell below 10% in 2004–2010, Turkish clubs achieved their best sustained European performances.
- Historical peaks — Galatasaray's 2000 UEFA Cup, Beşiktaş's 2017–18 UCL group stage, Turkey's 2002 World Cup 3rd place — align quantitatively with periods of relative economic stabilisation. The data formally supports this narrative.

England — Economy Decoupled

- England's Big 6 clubs show a negative (non-significant for H_1) correlation with domestic EHI. Premier League broadcast revenue and foreign club ownership insulate European performance from domestic economic conditions.
- The Abu Dhabi acquisition of Manchester City in August 2008 — at the onset of the Global Financial Crisis — is the most visible example: a period of UK economic

weakness catalysed a wave of foreign investment that transformed English club competitiveness.

- England's national team does show a small but significant positive EHI correlation ($r = +0.240$, $p = 0.028$), suggesting economic conditions influence grassroots development even when top club performance is insulated.

Machine Learning Summary

- Turkey alone ($n=64$): 68.9% CV accuracy vs. 60.9% baseline — modest but meaningful improvement from economic features.
- England alone ($n=64$): 86.0% accuracy, partly driven by temporal structure rather than pure economic signal.
- Combined ($n=128$): 90.7% accuracy (Random Forest) — combining two structurally distinct countries amplifies discriminative power.
- Country identification: 100% accuracy — Turkey and England are completely separable from economic and football features alone.
- K-Means ($k=3$): Three natural clusters map to Turkey's three economic-footballing eras (hyperinflation 1990s, stability 2000s, mixed 2010s–2020s).

9. Limitations & Future Work

Current Limitations

- Sample size: $N = 64$ per country is below the ideal for machine learning with 5 features. The combined $N = 128$ partially addresses this, but per-country models remain underpowered for complex non-linear classifiers.
- Temporal resolution: GDP and CPI are annual figures assigned to both half-years of the same year, artificially reducing temporal granularity. Monthly economic data would allow a more precise alignment with the UEFA season calendar.
- Data reconstruction: Per-club half-year UEFA coefficient points were reconstructed from archival sources. While cross-validated against multiple databases, some historical values (particularly 1993–2000) have lower confidence than recent seasons.
- Omitted confounders: The model does not capture manager quality, transfer policy decisions, club boardroom stability, player injury records, or youth academy investment. These factors explain a substantial portion of the unexplained 39% variance in the OLS model.
- Causality: All results are correlational. Higher inflation may reduce club UEFA performance, or both may be jointly driven by a third variable (e.g. political instability causing both economic deterioration and reduced government support for football infrastructure).
- England's EHI measurement: Using GBP/USD as the exchange rate component for England assumes a dollar-denominated player market, which is true for transfer fees but

less so for wages. A euro-denominated rate would be more appropriate for European football economics.

Future Work

- Transfermarkt squad values: Incorporating half-year squad market values (in euros) would add a direct monetary measure of club investment quality, bridging macroeconomics and football-specific financial data.
 - Multi-country panel analysis: Extending to Germany (Bundesliga), Spain (La Liga), France (Ligue 1), and Italy (Serie A) would create a true panel dataset ($N > 400$) enabling fixed-effects regression and much stronger causal inference.
 - NLP on transfer market commentary: Sentiment analysis of sports journalism and club financial reports could capture qualitative economic signals not captured by GDP/CPI.
 - Non-linear economic thresholds: The relationship between inflation and club performance may not be linear — a threshold model (e.g. above 20% inflation, the effect is catastrophic; below 10%, it plateaus) might better capture the data.
 - Post-2024 extension: The 2022–2024 period represents a new high-inflation crisis for Turkey. Extending the dataset will provide a natural out-of-sample test of the model's predictive validity.
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10. References

Data Sources

- USD/TRY exchange rate: Central Bank of Turkey (TCMB Electronic Data Delivery System) — evds2.tcmb.gov.tr
- GBP/USD exchange rate: Bank of England Statistical Database — bankofengland.co.uk
- GDP growth & CPI inflation (Turkey & England): World Bank Open Data — data.worldbank.org (indicators NY.GDP.MKTP.KD.ZG and FP.CPI.TOTL.ZG)
- CPI inflation supplement (Turkey): TurkStat (Turkish Statistical Institute) — tuik.gov.tr
- CPI inflation supplement (England): ONS (Office for National Statistics) — ons.gov.uk
- UEFA club coefficients: UEFA.com / rsssf.org / kassiesa.net/uefa/coeff
- FIFA World Rankings (historical): FIFA Historical Archive — fifa.com/fifa-world-ranking

Python Libraries

- pandas, numpy — data wrangling, time series alignment, feature engineering
- matplotlib, seaborn — all visualisations (Figures 1–12)
- scipy.stats — Pearson r , Spearman ρ , D'Agostino K^2 normality test, Fisher CI

- statsmodels — OLS regression, Breusch-Pagan test, Durbin-Watson statistic
- scikit-learn — Logistic Regression, Random Forest, Gradient Boosting, K-Means, PCA, StratifiedKfold

Code & Documentation

- GitHub repository: github.com/MuhammedGunes54/DSA-210-Project
- Project webpage: muhammedgunes54.github.io/DSA-210-Project
- Analysis script: `analysis_v3.py`
- Dataset: `master_data_v2.xlsx`
- Requirements: `requirements.txt`
- AI usage disclosure: `AI_USAGE.md`

11. AI Tool Disclosure

This project used Claude (Anthropic, claude.ai) as an AI assistant for four specific tasks. All analytical decisions, hypothesis framing, interpretation of results, and written conclusions were made by the student. Full disclosure including specific prompts and outputs is documented in `AI_USAGE.md` in the GitHub repository, as required by DSA 210 academic integrity guidelines ("Use of AI tools is permitted but must be explicitly documented, including specific prompts used and outputs generated.").

Task	Prompt (abbreviated)	Output Used	Student Contribution
Code refactoring	'Refactor my script into functions with PEP8 style. Don't change logic or outputs.'	Function structure, docstrings, variable names in <code>analysis_v3.py</code>	All statistical logic, model selection, output interpretation
Economic index formula	'Combine USD/TRY, GDP, CPI into a 0–1 index. Weaker lira and higher inflation should score lower.'	Min-max normalisation + inversion formula (adopted as-is)	Decision to use it; verification against economic literature
Dataset schema	'Give me column structure for half-year economic-football data for TR and ENG.'	Column names and 5 example rows for <code>master_data_v2.xlsx</code>	All data values (manually sourced from primary databases)
Webpage HTML/CSS	'Build a dark-theme single-page HTML report. Navy, red, teal, gold palette. Bebas Neue headings.'	Full HTML/CSS layout structure for <code>index.html</code>	All section text, all findings, all analysis narrative

All p-values, correlation coefficients, R^2 , F-statistics, and ML accuracy figures were produced by running analysis_v3.py locally. No AI tool was used to generate, fabricate, or interpret numerical results.